

Key Competencies & Knowledge Areas

- **Programming Languages:** Python (essential), with basics in libraries like NumPy, pandas, matplotlib, scikit-learn, TensorFlow/PyTorch
- **Mathematics:** Linear algebra, probability, statistics, calculus (core for understanding ML algorithms)
- **Data Manipulation & Cleaning:** Data preprocessing, feature engineering
- **Data Visualization:** Using seaborn, matplotlib for exploratory analysis
- **Machine Learning Algorithms:** Supervised (regression, classification), unsupervised (clustering, dimensionality reduction), ensemble learning
- **Model Evaluation & Validation:** Metrics, splitting data, cross-validation, avoiding overfitting/underfitting
- **Advanced Topics:** Deep learning (neural networks), NLP, computer vision, time series
- **Deployment:** Packaging ML models as APIs or web services, using cloud ML platforms (optional for advanced)
- **Tools:** Jupyter Notebooks, Git/GitHub, Google Colab, Kaggle

Sequenced Learning Milestones

Short-Term (Months 1–3): Foundations

1. Python Programming for ML

○ Resources:

- [Automate the Boring Stuff with Python](#)
- [Kaggle Python Course](#)

- **Project:** Data exploration and manipulation using a real-world dataset (e.g., Titanic, Iris).

2. Math Foundations (Linear Algebra, Probability, Statistics)

○ Resources:

- [Khan Academy Math for Data Science](#)

- [Essence of Linear Algebra \(YouTube\)](#)
- **Project:** Write and visualize your own implementation of linear regression from scratch in Python.

3. Data Manipulation & Visualization

- **Resources:**
 - [pandas Official Documentation](#)
 - [Data Visualization with Python \(Coursera\)](#)
- **Project:** Perform exploratory data analysis (EDA) on a Kaggle dataset; visualize correlations and insights.

Mid-Term (Months 4–7): Core Machine Learning

4. Basic Supervised & Unsupervised Learning Algorithms

- **Resources:**
 - [Kaggle Intro to Machine Learning](#)
 - [scikit-learn User Guide](#)
- **Project:** Implement classification (e.g., cancer detection) and clustering (e.g., customer segmentation) on real datasets.

5. Model Evaluation, Validation, and Selection

- **Resources:**
 - [Coursera: Machine Learning – Andrew Ng](#)
 - [Practical Machine Learning \(Book\)](#)
- **Project:** Tune and compare several algorithms on one dataset; create a report on their strengths and weaknesses.

6. Feature Engineering and Data Pipelines

- **Resources:**
 - [Feature Engineering \(Kaggle\)](#)
 - [Hands-On Machine Learning with Scikit-Learn \(Book\)](#)

- **Project:** Build an automated pipeline from raw to cleaned data, feature creation, and model building using scikit-learn's Pipeline.

Long-Term (Months 8–12): Advanced Topics & Deployment

7. Deep Learning Fundamentals

- **Resources:**
 - [Deep Learning Specialization \(Andrew Ng, Coursera\)](#)
 - [TensorFlow/Keras Docs](#)
- **Project:** Use Keras/TensorFlow or PyTorch to build and train a basic image or text classification neural network.

8. Special Topics: NLP, Computer Vision, Time Series (Pick based on interest)

- **Resources:**
 - [Natural Language Processing with Python \(NLTK Book\)](#)
 - [FastAI for Vision/NLP](#)
- **Project:** Sentiment analysis on tweets, facial recognition, or forecasting with time series data.

9. Model Deployment & MLOps Basics

- **Resources:**
 - [Flask REST API Tutorial](#)
 - [TensorFlow Serving / Heroku Deployment Guide](#)
- **Project:** Package your ML model as an API and deploy it via Flask or Streamlit so others can use it from a web interface.

10. Portfolio & Communication

- **Resources:**
 - [GitHub Pages](#)
- **Project:** Assemble a portfolio of all completed projects with READMEs, visualizations, and demo links.

Step-by-Step Structure

Step 1: Gain Python & Math Foundations

- Daily coding and math exercises.
- Complete one project weekly (small analysis or algorithm from scratch).

Step 2: Master Data Handling & Visualization

- Practice manipulating and visualizing at least three real datasets.
- Document insights and EDA in Jupyter Notebooks.

Step 3: Core ML Modeling

- Complete at least three ML projects covering regression, classification, and clustering.
- Compare and choose the best models for each.

Step 4: Advance to Deep Learning and Special Topics

- Take foundational deep learning courses.
- Implement a basic neural net; experiment with image or text data.

Step 5: Deployment & MLOps Basics

- Learn to serve models with Flask/Streamlit.
- Deploy at least one ML service publicly; present demos in your portfolio.

Step 6: Portfolio & Continuous Growth

- Add new case studies and document lessons monthly.
- Apply to ML-focused problems at work or in open competitions (Kaggle, DrivenData).

Short-Term Goals (Months 1-6):

- Master Python and ML math fundamentals.
- Complete 3+ hands-on ML projects (classic algorithms).

- Build proficiency in scikit-learn and pandas.

Long-Term Goals (Months 7–12):

- Implement deep learning models (Keras/TensorFlow/PyTorch).
- Deploy at least one ML app.
- Build a professional portfolio (GitHub, blog, or personal site) with 5+ projects and clear technical write-ups.
- Be ready for junior or associate ML engineer/Data Scientist interviews.

Progress Tracking:

- **Weekly:** Log time spent learning, and solve one small problem set.
- **Monthly:** Finish a project or Kaggle competition and review lessons learned.
- **Quarterly:** Publicly share your work, request feedback, and re-align goals based on interests and job needs.

Following this roadmap will provide systematic, hands-on advancement in machine learning, ensuring both depth and breadth for professional success.